DATE: OCTOBER 13, 1999

EVAPCO SUBMITTAL PACKAGE

PROJECT BUREAU OF ENGRAVING & PRINTING UNIT (1) AT 424-936 COOLING TOWER

CUSTOMER AMERICAN COMBUSTION, INC. P.O. 6744

EVAPCO SERIAL NUMBER 997139M ENGINEER WASHINGTON GAS ENERGY SYSTEMS

SUBMITTAL DATA ENCLOSED

UNIT CERTIFIED DRAWINGS

THERMAL PERFORMANCE GUARANTEE

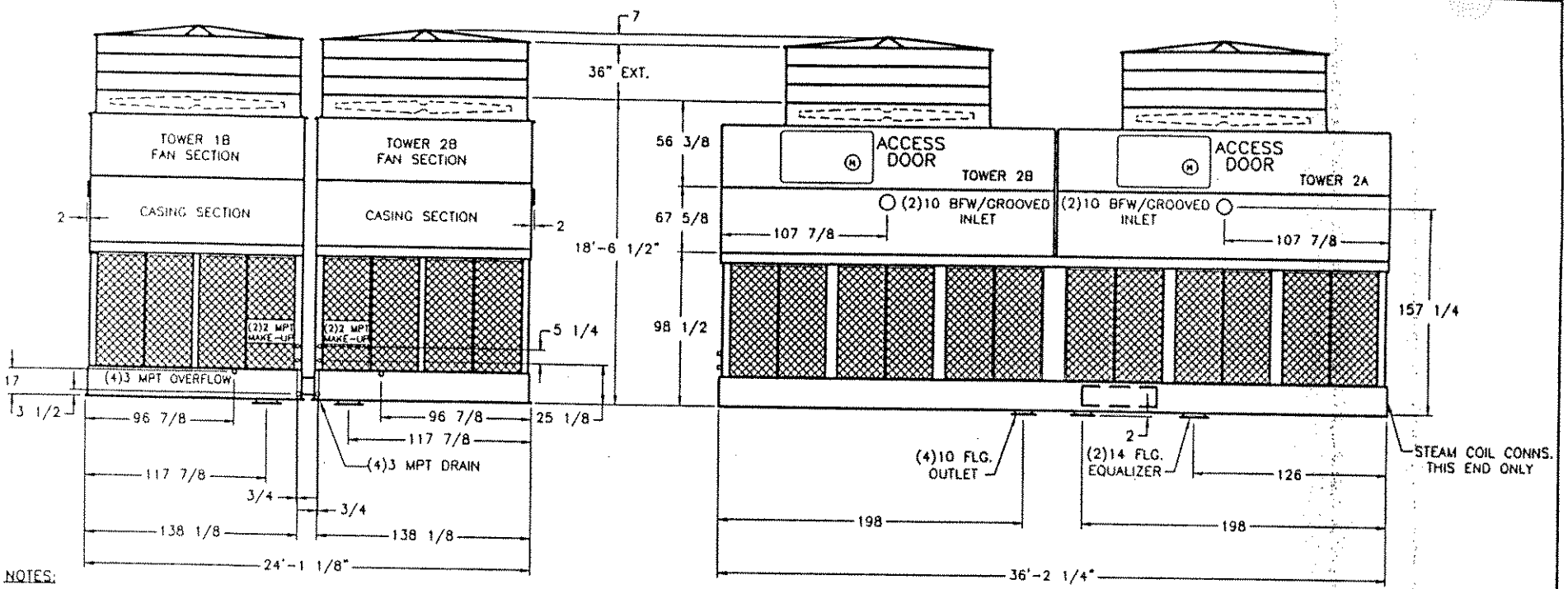
RECOMMENDED STEEL SUPPORT DRAWINGS

MECHANICAL SPECIFICATIONS

WARRANTY STATEMENT

ACCESSORY INFORMATION CHECKED "✓" WHEN INCLUDED

- Electric Damper Controls
- Electric Pan Heater
- ✓ Electric Water Level Control
- Discharge Hood
- Discharge Hood With Dampers
- Vibration Rails
- ✓ Vibration Cut-Out Switch
- Two-Speed Motor Wiring Diagram
- Pony Motor
- ✓ External Service Platform
- Ladder (if not shown on the certified print)
- Inlet Sound Attenuation Drawing
- Outlet Sound Attenuation Drawing
- ✓ Hot Water/Steam Coil General Arrangement



NOTES:

1. Ⓜ -FAN MOTOR LOCATION
2. MAKE-UP WATER PRESSURE 25psi MIN. 125psi MAX.
3. 3/4" DIA. MOUNTING HOLES. REFER TO RECOMMENDED STEEL SUPPORT DRAWING.
4. HEAVIEST SECTION IS LOWER SECTION. WEIGHT INCLUDES 1670 lb. REMOVABLE SKID.
5. BFW DENOTES BEVELED FOR WELDING.
   FPT DENOTES FEMALE PIPE THREADS.
   MPT DENOTES MALE PIPE THREADS.
6. FAN SECTION WEIGHT IS 3730 lbs. EACH. THERE ARE (4)FAN SECTIONS FOR THIS UNIT.
   CASING SECTION WEIGHT IS 5560 lbs. EACH. THERE ARE (4)CASING SECTIONS FOR THIS UNIT.
   SECTION WEIGHT INCLUDES 810 lb. SKID.
   PAN SECTION WEIGHT IS 7180 lbs. EACH. THERE ARE (2)PAN SECTIONS FOR THIS UNIT.
   SECTION WEIGHT INCLUDES 1670 lb. SKID.
7. FAN AND CASING SECTIONS WILL BE SHIPPED WITHOUT SEALER TAPE BETWEEN SECTION JOINT. SECTIONS MUST BE UNBOLTED PRIOR TO LIFTING. SEALER TAPE TO BE INSTALLED DURING RIGGING.
8. FAN CYLINDERS WILL BE SHIPPED LOOSE FOR FIELD MOUNTING. EACH SECTION WILL BE 3'-7" TALL.

WEIGHTS (LB)			NO. SHIPPING SECTIONS	NO. RIGGING SECTIONS
SHIPPING	OPER.	HEAVIEST SECTION		
51520	88800	7180	10	14

MODEL NUMBER: AT 424-936 COOLING TOWER TAG: TOWER 1 & 2

CERTIFIED FOR AMERICAN COMBUSTION, INC. PROJECT: BUREAU OF ENGRAVING & PRINTING

CUSTOMER ORDER NO. 6744 EVAPCO NO. 997139M

CAPACITY 8,250 G.P.M. 95 °F IN 85 °F OUT 78 °F E.W.B.

FAN MOTOR HP (4) 60 ELEC. SPEC. 460/60/3

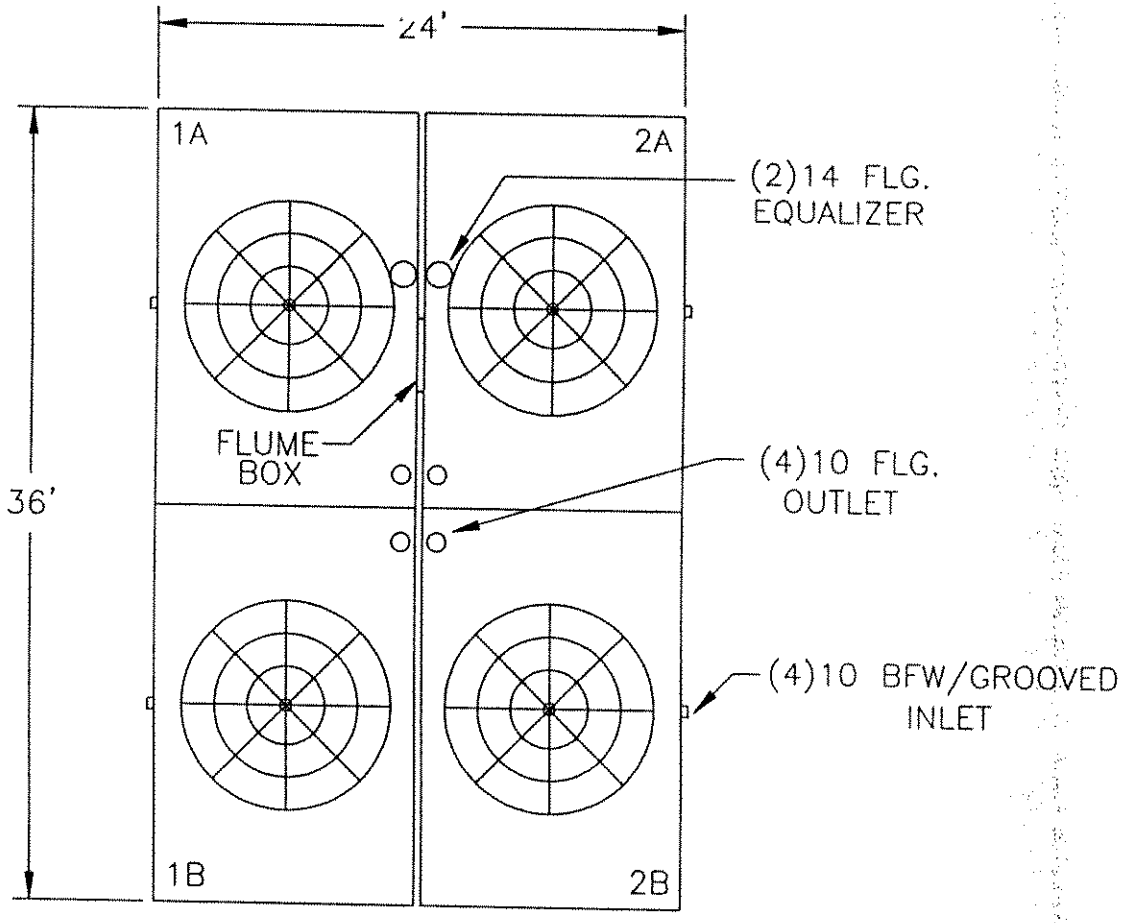
INLET PRESSURE 2.7 P.S.I.G. DRIVES SIZED FOR 0" ESP.

REMARKS: UNIT FURNISHED WITH 5-PROBE ELECTRIC WATER LEVEL CONTROL PACKAGE, (4) EXTERNAL SERVICE PLATFORMS WITH (4) LADDERS & SAFETY CAGES, (4) VIBRATION CUTOUT SWITCHES, STAINLESS STEEL STEAM HEATER COILS, 3' STAINLESS STEEL FAN CYLINDER EXTENSION (TO SHIP ASSEMBLED WITH FAN SCREENS ATTACHED, ON SEPARATE TRUCK), (2) 14" FLANGED EQUALIZER CONNECTIONS, (4) 10" FLANGED OUTLET CONNECTIONS AND ALL STAINLESS STEEL CONSTRUCTION PER SPECIFICATION SHEET. NOTE: STENCIL UNIT LEFT HAND SECTION 1A & 1B, AND RIGHT HAND SECTION 2A & 2B.

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COOLING TOWER

T2243648ERA-05



PLAN VIEW

TOWER 1/2

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T22436-01

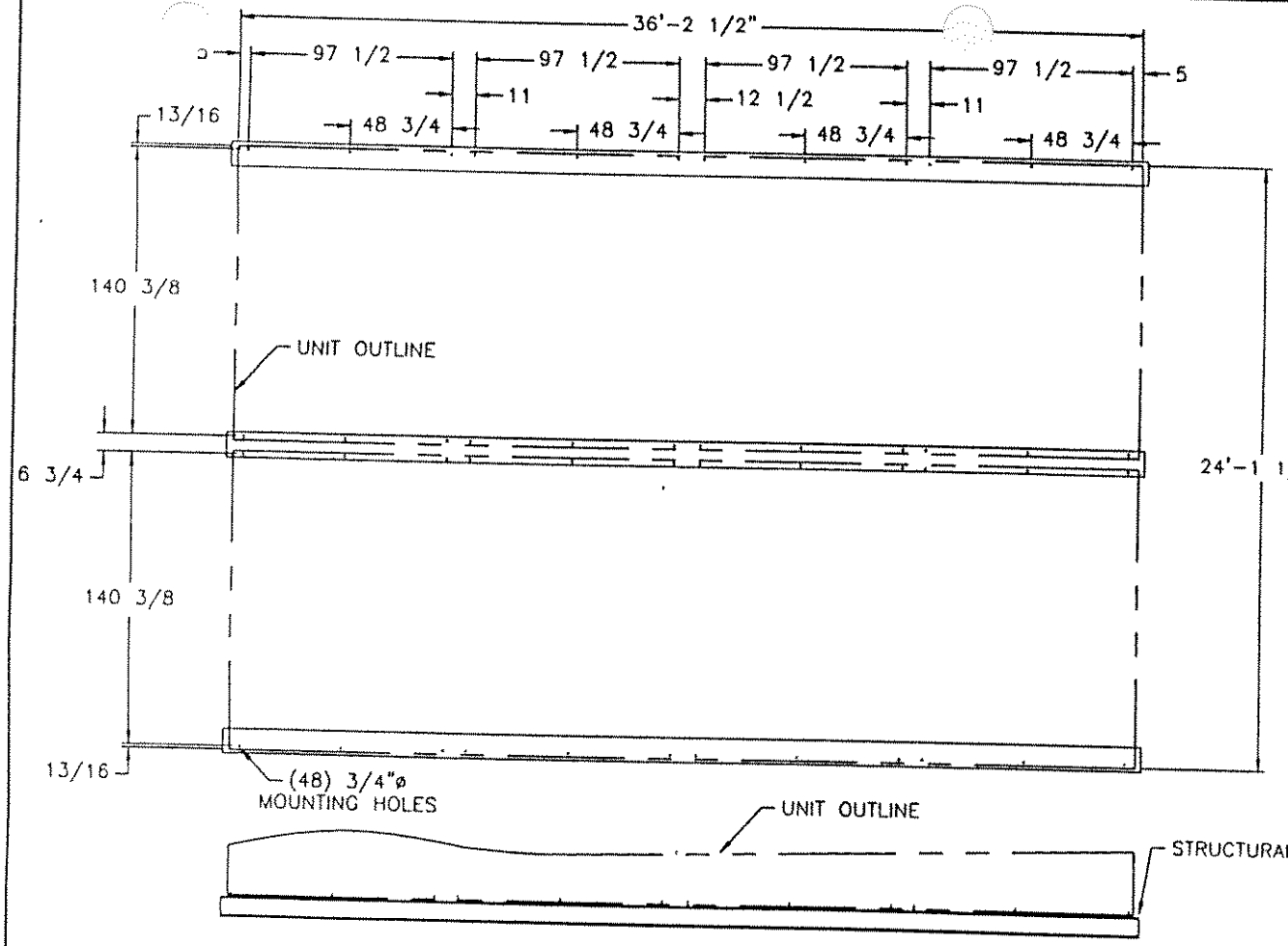
Guarantee of Thermal Performance by EVAPCO

EVAPCO® unequivocally guarantees the thermal performance of its equipment as shown on the certified drawings, when the equipment is installed in accordance with good engineering practice. If after installation and start-up there is any question regarding thermal performance of the equipment, at the owner's request EVAPCO will send its engineers to the jobsite to conduct a performance test. This test may be observed by the owner and the consulting engineer or by their authorized representatives. If the results of the evaluation show the equipment to be deficient, EVAPCO will make the necessary repairs or alterations to correct the deficiency at no cost to the owner. If the equipment is found to be performing in accordance with its certified drawing, the owner is expected to reimburse the company for its costs associated with this performance test.

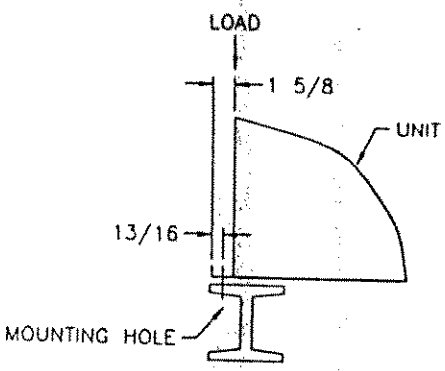
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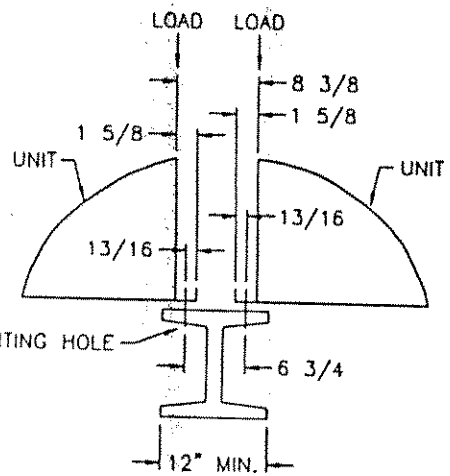
OH5000/499



ALL 24x36 INDUCED DRAFT UNITS



TYPICAL END VIEW



CENTER BEAM ARRANGEMENT

NOTES:

1. BEAMS SHOULD BE SIZED IN ACCORDANCE WITH ACCEPTED STRUCTURAL PRACTICES. MAXIMUM DEFLECTION OF BEAM UNDER UNIT TO BE 1/360 OF UNIT LENGTH NOT TO EXCEED 1/2".
2. DEFLECTION MAY BE CALCULATED BY USING 56% OF THE OPERATING WEIGHT ON THE CENTER BEAM AND 22% ON EACH OUTBOARD BEAM. SEE CERTIFIED PRINT FOR OPERATING WEIGHT.
3. THE CENTER BEAM SHOULD HAVE A MINIMUM WIDTH OF 12".
4. SUPPORT BEAMS AND ANCHOR BOLTS ARE TO BE FURNISHED BY OTHERS.
5. BEAMS MUST BE LOCATED UNDER THE FULL LENGTH OF THE PAN SECTION.
6. SUPPORTING BEAM SURFACE MUST BE LEVEL. DO NOT LEVEL THE UNIT BY PLACING SHIMS BETWEEN THE UNIT MOUNTING FLANGE AND THE SUPPORTING BEAM.
7. ALL 24x36 MODELS ARE MULTIPLE CELL UNITS. OPERATING WEIGHT OF EACH CELL IS FOUND BY DIVIDING TOTAL OPERATING WEIGHT BY THE NUMBER OF CELLS.

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RECOMMENDED STEEL SUPPORT

T22436ERA-SL

DATE OCTOBER 13, 1999

MECHANICAL SPECIFICATIONS

EVAPCO® AT COOLING TOWERS

PROJECT BUREAU OF ENGRAVING & PRINTING UNIT (1) AT 424-936 COOLING TOWER

CUSTOMER AMERICAN COMBUSTION, INC. P.O. 6744

EVAPCO SERIAL NUMBER 997139M ENGINEER WASHINGTON GAS ENERGY SYSTEMS, INC. – SPRINGFIELD, VA

UNIT TYPE	Factory assembled, induced draft, counterflow cooling tower.
CONSTRUCTION	Type 304 Stainless steel casing and pan with Type 304 Stainless steel channel and angle supports. Fan cowl is constructed of stainless steel.
MAKE-UP FLOAT VALVE ASSEMBLY*	Brass float valve with adjustable plastic float.
PAN STRAINER*	All type 304 stainless steel construction with large area removable perforated screens.
ACCESS	Hinge mounted door in the upper casing for fan drive and water distribution system access. Removable louver panels on all four sides of the unit for pan and sump access.
NS	Fans are axial propeller type constructed of aluminum alloy and statically balanced. The fan is installed in a closely fitted cowl with venturi air inlet. Fan screens are galvanized steel and have steel frames bolted to the fan cowl.
FAN SHAFT	Solid shaft of ground and polished steel. Exposed surface coated with rust preventative.
FAN SHAFT BEARINGS	Heavy-duty, self-aligning ball type bearings with extended lubrication lines to grease fittings located on access door frame. Bearings are designed for a minimum L-10 life of 75,000 hours.
FAN MOTOR	Totally enclosed, ball bearing type electric motor(s) suitable for moist air service. Motor(s) are 1.15 service factor design.
FAN DRIVE	The fan drive is a multi-groove, solid back, reinforced neoprene V-belt type with taper lock sheaves designed for 150% of the motor nameplate horsepower. Fan and motor sheaves are constructed of aluminum alloy.
FILL	Polyvinyl Chloride (PVC) of cross-fluted design. PVC sheets are bonded together for strength and durability. Fill is self-extinguishing for fire resistance, has a flame spread of 5 under A.S.T.M. designation E-84-81a, and is resistant to rot, decay and biological attack.
WATER DISTRIBUTION SYSTEMS	Precision molded ABS spray nozzles with large 3/8" x 1" orifice to eliminate clogging. Spray header and branches are Polyvinyl Chloride (PVC) for corrosion resistance with steel connection to attach external piping. Branches have threaded end caps to assist with debris removal.
ELIMINATORS	The eliminators are constructed entirely of Polyvinyl Chloride (PVC) in easily handled sections. Design incorporates three changes in air direction and limits the water carryover to a maximum of 0.001% of the circulating water rate.
LOUVERS	The louvers are constructed from Polyvinyl Chloride (PVC) and are mounted in easily removable frames for access to the pan for maintenance. There are two changes in air direction to prevent splash-out and block direct sunlight.

* OMITTED ON UNITS FOR REMOTE SUMP OPERATION All SST 12 FT WIDE BELT DRIVE 2801

Manufacturers of
Heat Transfer Equipment

evapco

EVAPCO, INC.

P. O. Box 1300
Westminster, Maryland 21158, USA

Telephone (410) 756-2600
FAX (410) 756-6450

Project Name: BUREAU OF ENGRAVING & PRINTING

Model Number: (1) AT 424-936 COOLING TOWER

Serial Number: 997139M

EVAPCO® STANDARD INDUCED DRAFT WARRANTY

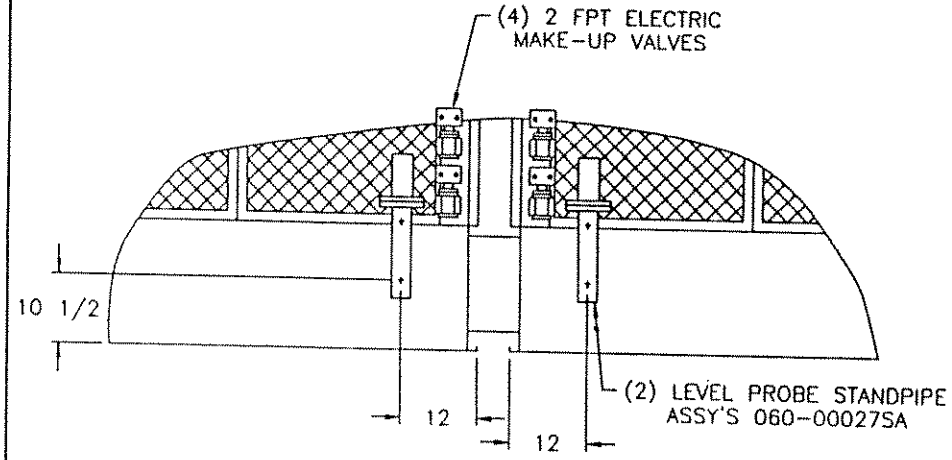
Five Year Total Product

This warranty applies to: AT, UBT, KT & REP Cooling Tower Products purchased with the stainless steel cold water basin and SST Cooling Tower Products.

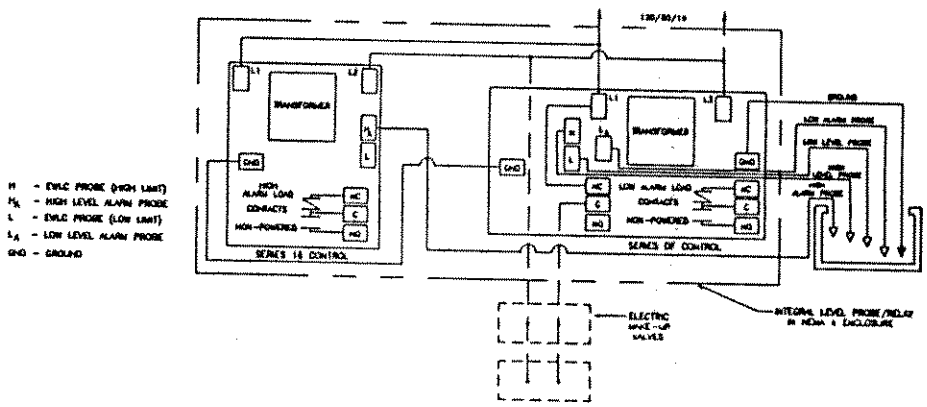
EVAPCO warrants that this product will be free from all defects in materials and workmanship for a period of **five (5) years** from the date of shipment by EVAPCO. This warranty includes all structural components, in addition to fill supports, drift eliminator supports, air inlet louvers and their supports and the fan discharge screen. The motor and drive system are covered by a comprehensive five (5) year warranty against defects in materials and workmanship. Included are the fan(s), bearings, pulleys, belts, shafts, gear reducer(s), drive shaft(s), drive couplings, electric fan motor(s) and mechanical equipment supports. All electrical components are covered for one (1) year from the date of start-up, but not to exceed eighteen (18) months from the date of shipment by EVAPCO. This warranty is predicated on the owner providing proper equipment maintenance as outlined in EVAPCO's Maintenance Instruction Bulletin.

In addition to the unit warranty above, EVAPCO warrants the thermal performance of the unit as shown on the certified drawings delivered to the customer for a period of one year from the date the installation is completed in accordance with good engineering practices, but in no event shall the warranty period exceed eighteen (18) months from the date the unit is shipped by EVAPCO. If after installation and start-up there is any question regarding thermal performance of the equipment, at the owner's request EVAPCO will send its engineers to the jobsite to conduct a performance test. This test may be observed by the owner and the consulting engineer or by their authorized representatives. If the results of the evaluation show the equipment to be deficient, EVAPCO will make the necessary repairs or alterations to correct the deficiency at no cost to the owner. If the equipment is found to be performing in accordance with its certified drawings, the owner will reimburse EVAPCO for all direct expenses incurred in connection with the performance test.

All other expenses or implied warranties of any kind whatsoever, including without limitation any warranties of merchantability or that the unit and equipment are fit for any particular purpose and any warranties that might otherwise arise out of a course of dealing between the parties or usage of trade, are hereby excluded. The sole remedy for breach of the warranties set out above shall be repair or replacement of the equipment by EVAPCO at its option. Under no circumstances shall EVAPCO be liable for lost profits, lost savings, personal injuries, incidental damages, economic loss, property damage, or other consequential, incidental, special or contingent damages. Even if EVAPCO has been advised of the possibility of such damages. In addition, EVAPCO shall not be responsible for any injuries or damages of any kind whatsoever under any theory of tort to the extent caused by misuse of the product by buyer or any third party.



PAN END VIEW
24x36 INDUCED DRAFT TOWERS



WIRING DIAGRAM

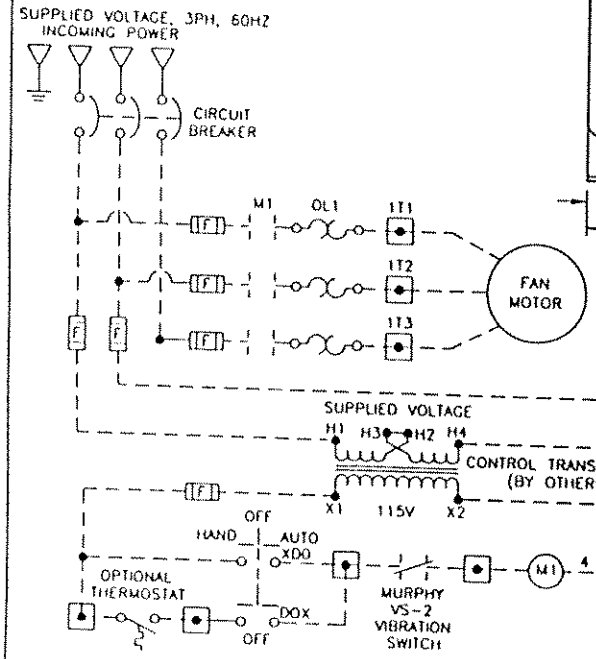
NOTES:
1. DASHED LINES INDICATE WIRING (BY OTHERS).
2. PIPING TO MAKE-UP VALVE (BY OTHERS).
3. LEVEL PROBE STANDPIPE ASSEMBLY AND MAKE-UP VALVE
   TO SHIP LOOSE FOR FIELD MOUNTING (BY OTHERS).
   TO SHIP MOUNTED.
4. SEE CERTIFIED PRINT FOR MAKE-UP LOCATION.
5. STANDPIPE TO BE HEAT TRACED AND INSULATED FOR WINTER OPERATION (BY OTHERS).
6. THE ELECTRONIC WATER LEVEL CONTROL ON THIS UNIT WILL MAINTAIN THE PROPER OPERATING WATER LEVEL. HOWEVER, BEFORE INITIAL START-UP THE UNIT MUST BE MANUALLY FILLED TO WITHIN 1" OF THE OVERFLOW.
7. VIEW SHOWN IS FOR A RIGHT HAND UNIT. LEFT HAND UNIT IS THE SAME VIEW SHOWN EXCEPT OPPOSITE END.

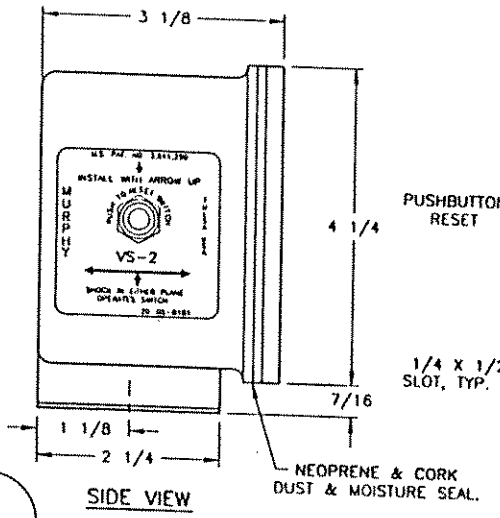
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ELECTRIC WATER LEVEL CONTROL WIRING & LOCATION

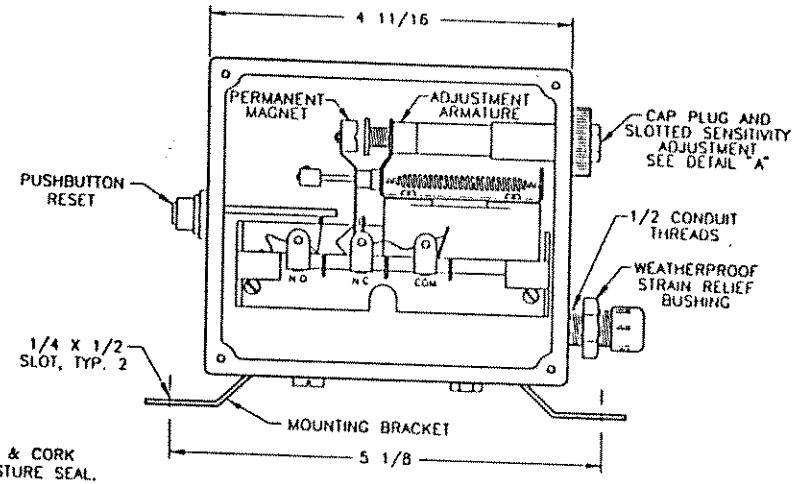
T22436ERA-01

NOTES:
1. DASHED LINES INDICATE WIRING(BY OTHERS)

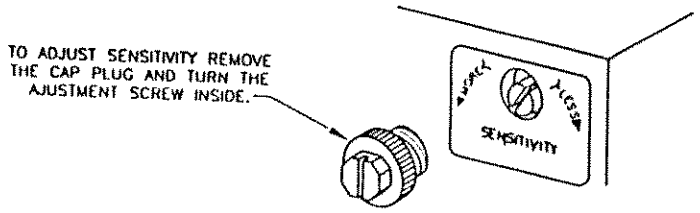




SIDE VIEW



FRONT VIEW
(COVER REMOVED)
MURPHY VS-2
VIBRATION SWITCH



DETAIL "A"

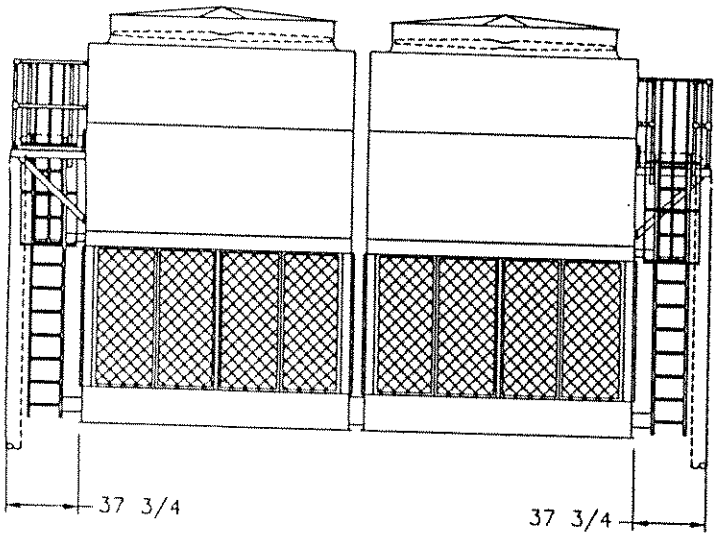
ADJUSTMENT

ADJUST THE SWITCH SO THAT DURING START-UP AND UNDER NORMAL CONDITIONS, THE CONTACTS DO NOT TRIP. FIRST, REMOVE THE PLASTIC PLUG AS SHOWN IN DETAIL "A" IN ORDER TO ACCESS THE ADJUSTMENT SCREW INSIDE. NEXT, WITH THE MOTOR OFF, TURN THE ADJUSTMENT SCREW COUNTER-CLOCKWISE (MORE SENSITIVE DIRECTION) UNTIL THE SWITCH TRIPS. NOW, TURN THE ADJUSTMENT SCREW CLOCKWISE 1/8 TURN (LESS SENSITIVE DIRECTION). RESET THE SWITCH BY DEPRESSING THE PUSH-BUTTON LOCATED UNDER THE RUBBER DUST COVER ON THE LEFT SIDE OF THE SWITCH. START THE MOTOR. IF THE MOTOR TRIPS THE SWITCH, THEN TURN THE ADJUSTMENT SCREW CLOCKWISE AN ADDITIONAL 1/8 TURN. RESET THE SWITCH AND START THE MOTOR AGAIN. REPEAT THE ABOVE PROCEDURE UNTIL THE MOTOR CONTINUES TO RUN.

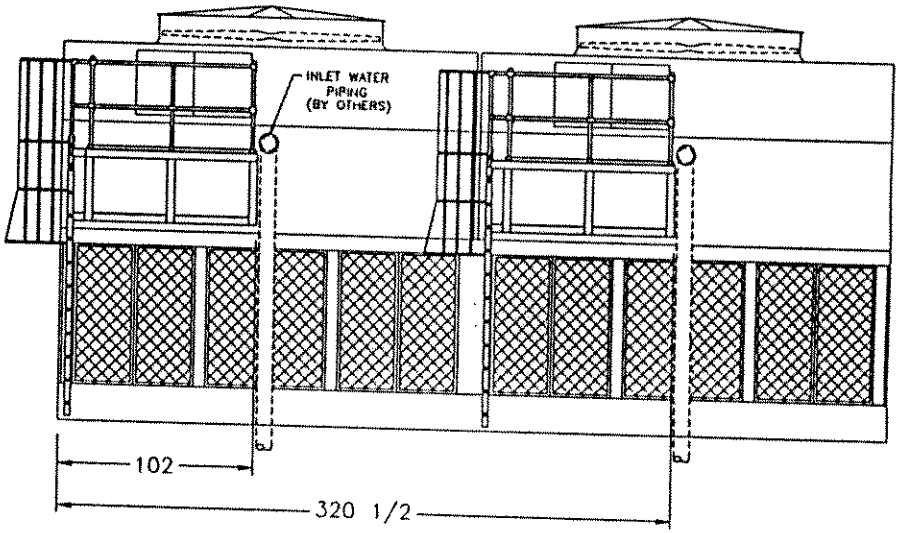
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MURPHY VS-2
VIBRATION SWITCH

VIB01ERB-WD



END VIEW



SIDE VIEW

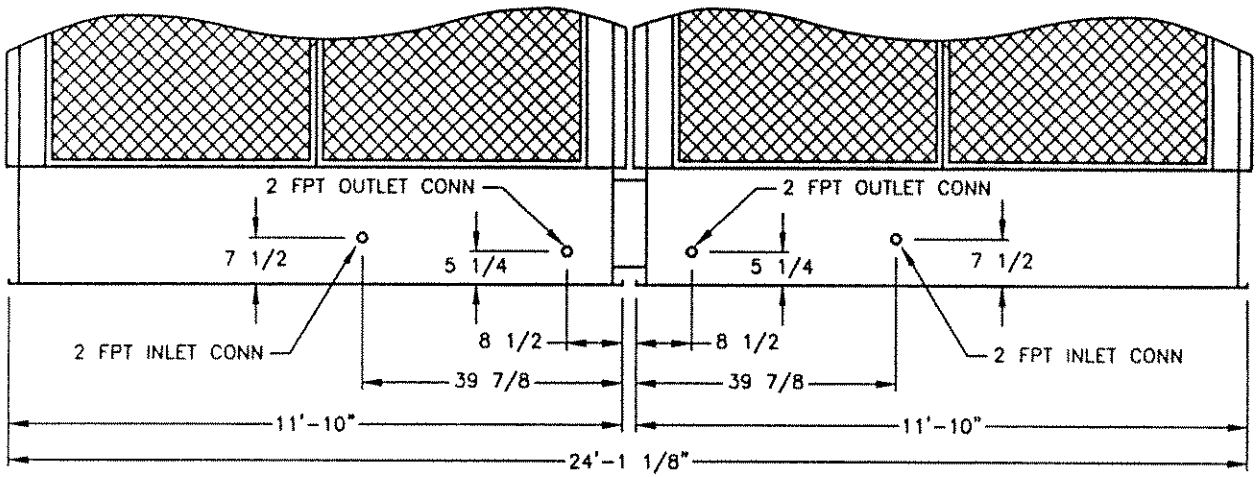
NOTES:

1. LADDER AND PLATFORM SHIP LOOSE. FIELD INSTALLATION BY OTHERS IS REQUIRED.
2. THE BOTTOM OF THE LADDER IS AT THE BASE OF THE UNIT. IF THE UNIT IS ELEVATED THEN AN OPTIONAL EXTENDED LADDER PACKAGE SHOULD BE CONSIDERED. (CONSULT FACTORY)
3. * REFER TO RIGGING PACK FOR LADDER AND PLATFORM MOUNTING INSTRUCTIONS.
4. Ⓜ – FAN MOTOR LOCATION.
5. MOUNT SERVICE PLATFORM BEFORE PIPING UNIT.

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EXTERNAL SERVICE PLATFORM

T22436ERA-02



END PANEL

24X24/36 INDUCED DRAFT TOWERS

NOTES:

1. FPT DENOTES FEMALE PIPE THREADS.
2. FPT CONNECTION IS A FULL COUPLING. IT PROTRUDES HALF IN AND HALF OUT.
3. UNIT CONNECTION NOT SHOWN FOR CLARITY.
4. STEAM COIL IS ONE CONTINUOUS CIRCUIT TYPICAL OF EACH CELL.

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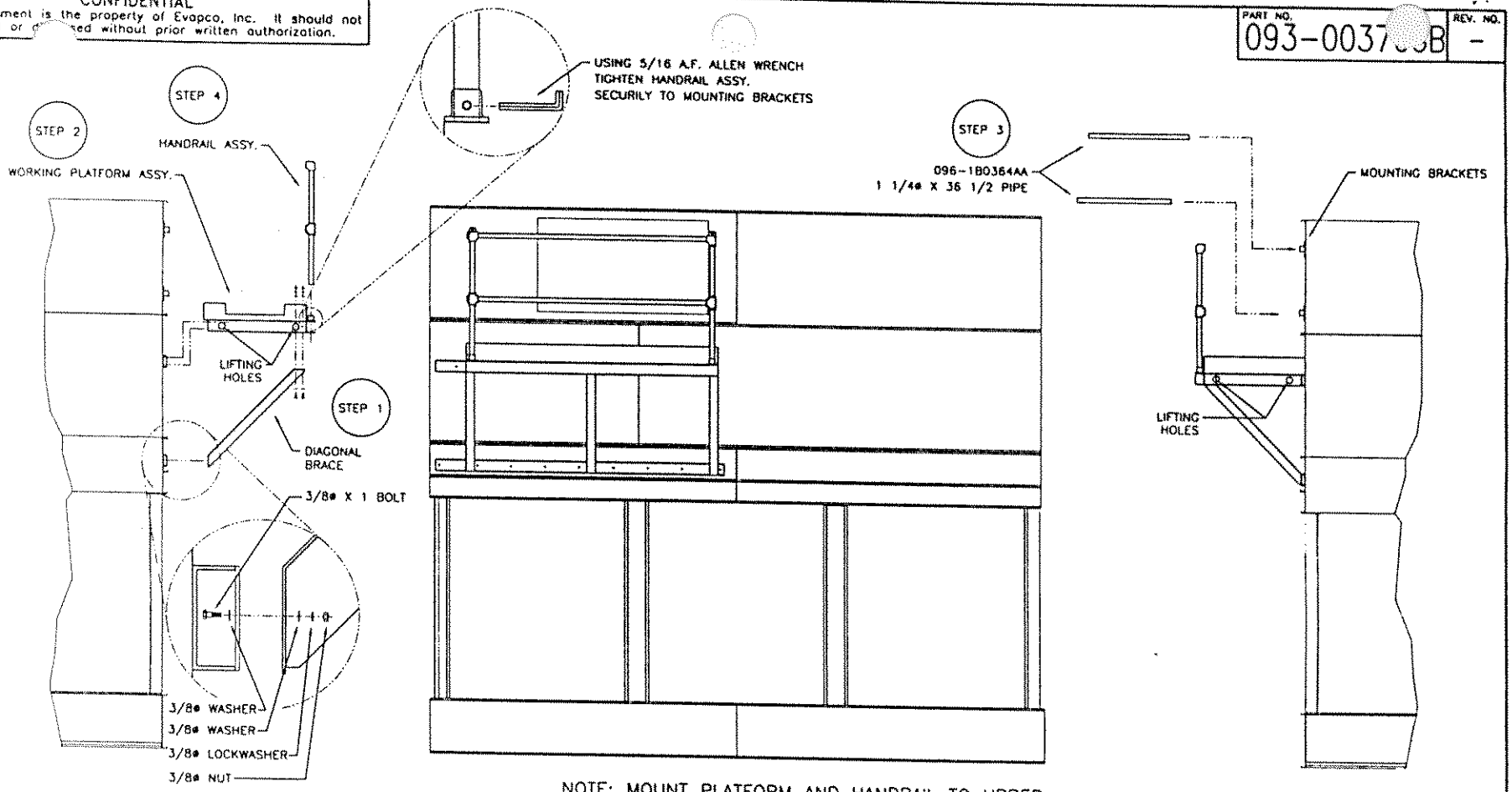
HOT WATER/STEAM COIL
GENERAL ARRANGEMENT

REV DATE

TA2424AB-A28

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PART NO. 093-00376GB REV. NO. –



FASTENER DETAIL (TYP.)

NOTE: MOUNT PLATFORM AND HANDRAIL TO UPPER SECTION BEFORE RIGGING TOP SECTION TO BOTTOM SECTION.

STAINLESS STEEL OPTION

PART NO.	RAW MAT'L

NOTE:
1. ALL 1/4ø HOLES SHOULD BE 11/32ø
2. USE STAINLESS STEEL N.C. SET-UP SHEET

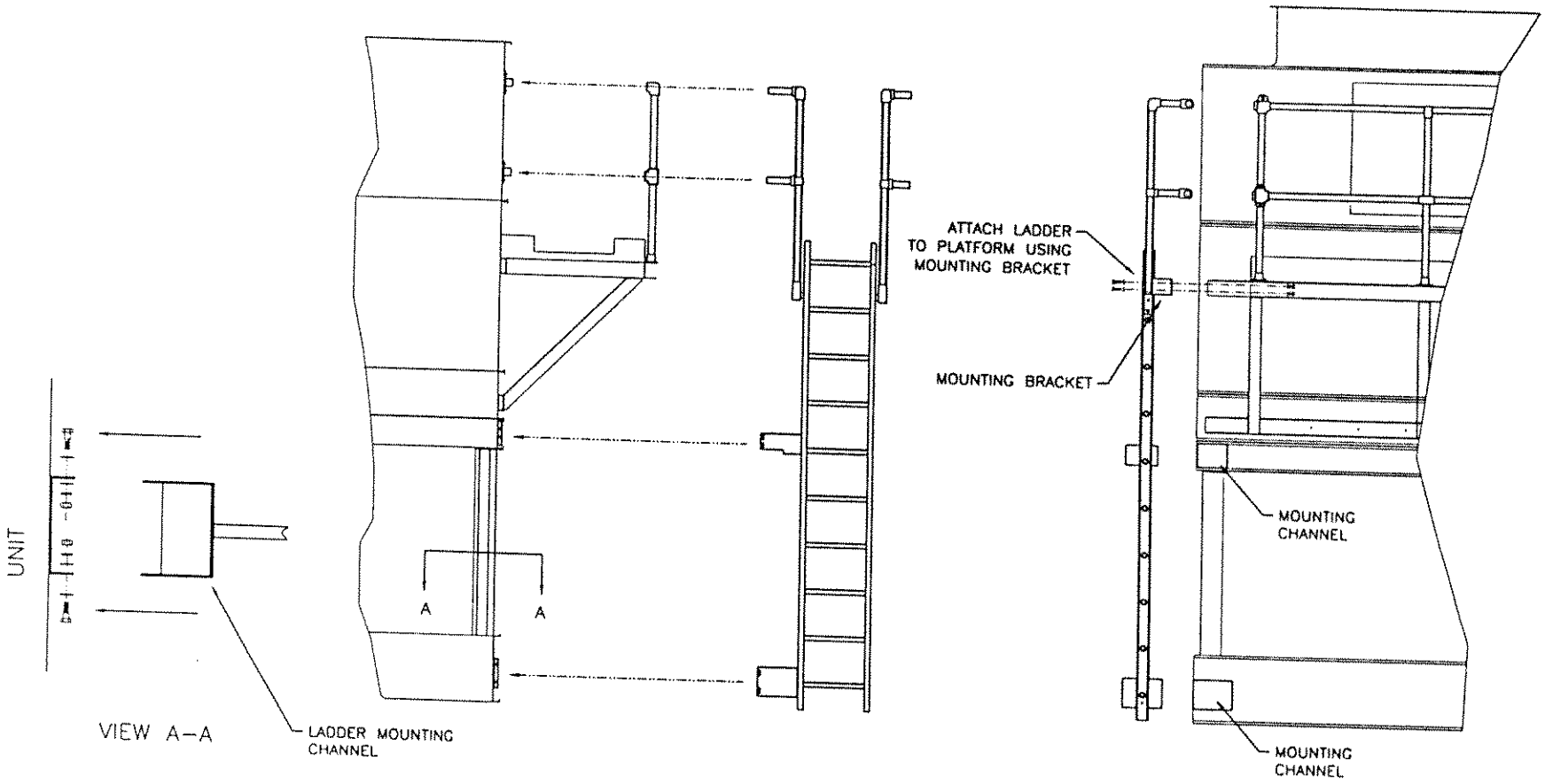
REVISIONS

EVAPCO, INC.

TITLE: PLATFORM FIELD ASSY.12X18 ATB		
CA	RAW MAT'L	CUT SIZE
DATE 5/9/97	SCALE N.T.S.	N.C. INFO.
DRAWN BY GO	CH'KD BY WS	PART NO. 093-00376GB REV. NO. –
NEXT ASSEMBLY:		

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PART NO. 093-00377GB REV. NO. -



STAINLESS STEEL OPTION

PART NO.	RAW MAT'L

NOTE:
1. ALL 1/4ø HOLES SHOULD BE 11/32ø
2. USE STAINLESS STEEL N.C. SET-UP SHEET

REVISIONS

EVAPCO, INC.

TITLE: LADDER MOUNTING ARR. 12X18		
CA	RAW MAT'L	CUT SIZE
DATE 5/10/97	SCALE N.T.S.	N.C. INFO.
DRAWN BY GO	CH'KD BY LvS	PART NO. 093-00377GB REV. NO. -
NEXT ASSEMBLY:		